

VIBRATION CONTROL USING MULTIPLE TUNED MASS DAMPERS

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This paper examines multiple tuned mass dampers (TMD's) with natural frequencies distributed over a frequency range. An integral form for the impedance is developed by asymptotic techniques. The result is used to analyze the vibration control capabilities of multiple TMD's for structures subjected to wide-band input. Calculus of variations is used to optimize the design of the TMD's with a constraint on the total mass. It is found that multiple TMD's can be more effective and more robust than a single TMD with equal total mass.

1. INTRODUCTION

A tuned mass damper (TMD) is a passive vibration control device consisting of a damped oscillator attached to a main structure. Den Hartog [1] developed analytical relations which demonstrate the vibration control capabilities of TMD's. For harmonic excitations, he derived optimal design parameters for such devices. Later works have generalized the theory to the control of structures subjected to wide-band excitation [2–4] and to the control of continuous systems [5–8].

Recently, systems with multiple TMD's have been proposed. Manikanahally and Crocker [8] considered systems in which each TMD is tuned to a different natural frequency of the structure. Fujino and his associates [9] have performed analytical and experimental investigations of large numbers of liquid oscillators attached to a main structure. In the latter study, all of the oscillators are tuned to the natural frequency of the main structure. The resulting multiple TMD system has the same dynamic properties as a single TMD of equal mass.

An alternate design principle is to use multiple oscillators with unequal natural frequencies distributed over a frequency range. It may seem that the oscillators with natural frequencies that are not exactly tuned to the natural frequency of the main structure are ineffective, implying that the design principle is inefficient. The purpose of this paper is to develop a theory to characterize multiple TMD's and to demonstrate that such devices can be more effective than a single TMD in reducing the response to wide-band input.

The basic system, shown in Figure 1, consists of n oscillators attached to a main structure, which is modelled by an oscillator. A wide-band excitation, modelled by white noise, is applied to the mass of the main oscillator, and the root-mean-square displacement response of that mass is of interest. The analysis begins with the impedance of multiple TMD's, defined as the force amplitude required to produce a unit harmonic velocity at the base of the TMD's. By Newton's law, this impedance is the sum of impedances of the

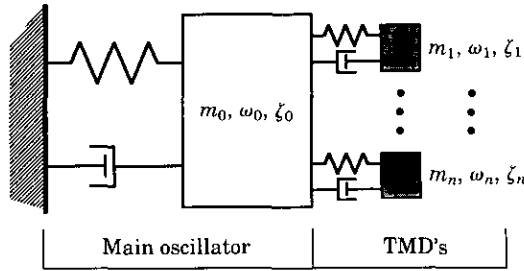


Figure 1. The system configuration.

individual TMD's. The authors have recently shown that, for a special case, the sum has simple limiting properties for increasing numbers of TMD's [10].

The first part of this paper extends these results to multiple TMD's with arbitrary properties. The analysis is based on concepts from Crandall's [11, 12] and Dowell and Kubota's [13] asymptotic theories and Skudrzyk's mean value theory [14, 15]. The only assumptions are that the natural frequencies of the TMD's are closely spaced and that the masses and damping ratios of the TMD's vary slowly with respect to their natural frequencies. The final asymptotic result for the main oscillator response is in terms of a simple integral.

The second part of this paper addresses the optimal design problem. The goal is to determine the properties of the multiple TMD's which minimize the response of the main oscillator. The constraint is given by an upper bound for the total mass of the TMD's. The approach used herein is based on the Euler-Lagrange equations from the calculus of variations [16]. This is a natural approach for the problem because of the integral form for the response. A closed form analytical solution is derived for the optimal design parameters.

In the optimal design, the multiple TMD's are not all exactly tuned to the main oscillator. This implies that multiple TMD's are more effective in vibration reduction than a single TMD with equal mass. An example illustrates this conclusion; moreover, it shows that multiple TMD's have robust properties under mistuned conditions: their performance degrades less than that of single TMD's when the natural frequency of the main oscillator is not equal to the value used in design.

2. ASYMPTOTIC ANALYSIS OF THE IMPEDANCE

Consider the system in Figure 1. The j th oscillator has natural frequency ω_j , damping ratio ζ_j , and mass m_j , for $j = 0, 1, \dots, n$, where $j = 0$ corresponds to the main oscillator and $j = 1, \dots, n$ correspond to the n TMD's. It is assumed that the natural frequencies of the TMD's are unequal and are ordered such that $\omega_1 < \omega_2 < \dots < \omega_n$.

The impedance $Z_j(\omega)$ of the j th TMD for a harmonic force of frequency ω applied at its base is

$$Z_j(\omega) = -i\omega m_j(\omega_j^2 - 2i\omega\omega_j\zeta_j)/(\omega_j^2 - \omega^2 - 2i\omega\omega_j\zeta_j), \quad (1)$$

where $i = \sqrt{-1}$. By Newton's law, the total impedance $Z(\omega)$ of the n TMD's is

$$Z(\omega) = \sum_{j=1}^n Z_j(\omega) = -i\omega \sum_{j=1}^n m_j(\omega_j^2 - 2i\omega\omega_j\zeta_j)/(\omega_j^2 - \omega^2 - 2i\omega\omega_j\zeta_j). \quad (2)$$

An earlier work [10] examined the special case in which the multiple TMD's have equal masses and damping ratios, and equally spaced natural frequencies. It was rigorously shown that, as n becomes large and the frequency spacing becomes sufficiently small, the impedance in equation (2) approaches a function that is slowly varying with respect to the excitation frequency, ω . In addition, it was found that the impedance is nearly independent of n if the total mass and the frequency range spanned by ω_j are held constant.

Here, the restriction on the multiple TMD's is relaxed to obtain new, generalized results for the impedance. The derivation consists of two basic steps: an asymptotic analysis and an integral limit. The asymptotic analysis is based on theories developed by Crandall [11, 12] and Dowell and Kubota [13], who found that many structures have modal properties with well-behaved asymptotic limits. In the present paper, a design condition is placed on the multiple TMD's so that their properties vary smoothly with respect to their natural frequencies. Then, asymptotic analysis is used to derive an integral expression for the impedance. This integral impedance result is found to be similar to those developed by Skudrzyk [14, 15] for rods, beams, plates and cylindrical shells. Skudrzyk applied a limit and used contour integration to derive simple forms for the integrals. The same techniques are used herein to obtain a final, simple expression for the impedance of the multiple TMD's.

The analysis of the impedance begins with some definitions. Define a non-dimensional frequency spacing parameter $\beta_j = (\omega_{j+1} - \omega_j)/\omega_j$ for $j = 1, \dots, n$. To make β_n well defined, it is necessary to assign a value for ω_{n+1} ; the value $\omega_{n+1} = \omega_n^2/\omega_{n-1}$ is used because this makes β_n equal to β_{n-1} . Also define functions $p(s)$ and $\zeta(s)$ for frequency s in the interval $\omega_1 \leq s \leq \omega_n$, such that they satisfy

$$p(\omega_j) = m_j/\beta_j \quad \text{and} \quad \zeta(\omega_j) = \zeta_j, \quad (3)$$

for $j = 1, \dots, n$. The function $p(s)$ represents the mass density of the multiple TMD's with respect to the normalized natural frequency. The impedance expression in equation (2) can be rewritten in term of $p(s)$ and $\zeta(s)$:

$$Z(\omega) = -i\omega \sum_{j=1}^n p(\omega_j) [\omega_j - 2i\omega\zeta(\omega_j)] \Delta\omega_j / (\omega_j^2 - \omega^2 - 2i\omega\omega_j\zeta(\omega_j)). \quad (4)$$

Here, $\Delta\omega_j = \omega_{j+1} - \omega_j$ for $j = 1, \dots, n$. The design condition on the multiple TMD's is that the functions $p(s)$ and $\zeta(s)$ vary smoothly with respect to the natural frequency argument, s . This condition permits asymptotic analysis of the impedance.

The asymptotic limit is defined as follows: the number of TMD's, n , goes to infinity while the total mass and the lowest and highest natural frequencies, ω_1 and ω_n , are fixed. Each value of n corresponds to a different set of multiple TMD's with new values for m_j , ζ_j , and ω_j . The relations in equation (3) are maintained for all n , and the frequency spacing becomes uniformly small, i.e., β_j satisfies $\lim_{n \rightarrow \infty} \max_j \beta_j \rightarrow 0$. Under these asymptotic conditions, it can be shown by the definition of the Riemann integral that the sum in equation (4) approaches

$$Z(\omega) \approx -i\omega \int_{\omega_1}^{\omega_n} [(s - 2i\omega\zeta(s))/(s^2 - \omega^2 - 2i\omega s\zeta(s))] p(s) s \, ds. \quad (5)$$

This integral is the exact asymptotic limit; however, an approximation sign is used because the number of TMD's is always finite. An error bound for the approximation was derived for the special case in which $p(\omega)$, $\beta(\omega)$ and $\zeta(\omega)$ are constant [17]. This error bound is briefly reviewed in the following.

Let e_m denote the maximum relative error of the integral approximation in equation (5). Then, the relationship between e_m and the parameters of the multiple TMD's was found to be [17]

$$\zeta > -(1/2\pi) \log(e_m/2\sqrt{4\pi^2 + 1})\beta. \quad (6)$$

This inequality gives a lower bound for the damping ζ , or, equivalently, an upper bound for the frequency spacing β . Thus, for an error tolerance $e_m = 0.1$, the damping and frequency spacing must satisfy $\zeta > 0.77\beta$. Numerical tests have shown that the error bound in equation (6) is also valid for multiple TMD's with more general properties, where $p(\omega)$, $\beta(\omega)$ and $\zeta(\omega)$ are smoothly varying functions of ω . The conclusion is that, under certain continuity conditions, the impedance of multiple TMD's with sufficiently closely spaced natural frequencies can be approximated by the integral in equation (5).

The integral can be further simplified using Skudrzyk's technique [14, 15] if the following additional conditions are satisfied: the damping of the oscillators is sufficiently small; the mass density function, $p(s)$, is slowly varying; and the frequencies ω_1 , ω_n and ω satisfy $\omega_1 \leq \omega \leq \omega_n$. First, the finite integral is approximated by a contour integral in the complex plane taken around the first quadrant. After the contour integral is evaluated by the residue theorem, the asymptotic limit is determined for $\zeta(s) \rightarrow 0$. The final approximate result is

$$Z(\omega) \approx (\pi/2)\omega p(\omega). \quad (7)$$

To illustrate the results of this section, consider 20 TMD's with natural frequencies spanning from $\omega_1/(2\pi) = 0.75$ Hz to $\omega_{20}/(2\pi) = 1.25$ Hz and a total mass of 10 kg. For simplicity, the TMD's are assumed to have equal masses, frequency spacing and damping ratios. To make the total mass equal to 10 kg, the individual TMD masses are $m_j = 0.5$ kg. To make the natural frequencies equally spaced, they must satisfy $\omega_j = \omega_1(1 + \beta_j)^{j+1}$. Since the ratio of the highest and lowest natural frequencies is $\omega_{20}/\omega_1 = 1.25/0.75 = (1 + \beta_j)^{19}$, the frequency spacing is $\beta_j = 0.0273$. To test the limits of the approximate results, the lower limit for the damping value, defined in equation (6), is used. For an error tolerance of $e_m = 0.1$, this corresponds to $\zeta_j = 0.0221$.

The impedance of the multiple TMD's is calculated using the exact expression (equation (2)) and the integral (equation (5)) and simple (equation (7)) approximations. The real and imaginary components are plotted in Figures 2(a) and (b), respectively. The plots show that the integral approximation is always accurate, and that the simple approximation has an accurate real component. The imaginary part of the simple approximation is reasonable only for a narrow range of frequencies midway between the lower and upper natural frequency limits. This limited accuracy of the simple approximation is sufficient for the design of multiple TMD's because the most significant frequencies in the response of the main oscillator are in a narrow band about the oscillator natural frequency.

3. VIBRATION REDUCTION PROPERTIES OF MULTIPLE TMD'S

The equation of motion of the main oscillator is given in terms of the reaction force $F_{\text{TMD}}(t)$ of the multiple TMD's:

$$m_0 \ddot{Y}(t) + 2m_0\omega_0\zeta_0 \dot{Y}(t) + m_0\omega_0^2 Y(t) = F(t) + F_{\text{TMD}}(t). \quad (8)$$

Here, $Y(t)$ is the displacement response of the main oscillator and $F(t)$ is the external force applied to the main oscillator mass. If the external force is a unit harmonic force, $F(t) = e^{-i\omega t}$, then the steady state response is of the form $Y(t) = y(\omega)e^{-i\omega t}$.

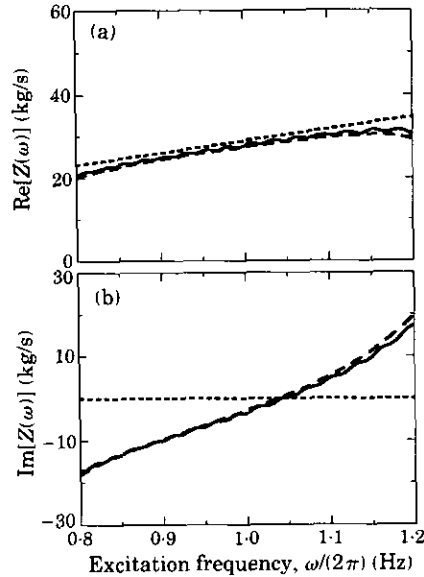


Figure 2. The impedance $Z(\omega)$ of 20 TMD's: (a) real part; (b) imaginary part. —, Exact; ---, integral approximation (equation (5)); - - -, simple approximation (equation (7)).

Also, by the definition of impedance, the reaction force of the TMD's is $F_{\text{TMD}}(t) = -i\omega y(\omega)Z(\omega)e^{-i\omega t}$. The complex amplitude of the harmonic response, $y(\omega)$, is obtained by substituting the preceding relations into equation (8):

$$y(\omega) = [m_0(\omega_0^2 - \omega^2 - 2i\omega_0\zeta_0) - i\omega Z(\omega)]^{-1}. \quad (9)$$

If the forcing function, $F(t)$, is a stationary process with one-sided power spectral density $G(\omega)$, then the mean square of the response of the main oscillator is [18]

$$\sigma_y^2 = \int_0^\infty G(\omega)|y(\omega)|^2 d\omega, \quad (10)$$

where $y(\omega)$ is obtained from equation (9).

To illustrate the effect of multiple TMD's on the harmonic response of the main oscillator, the same TMD's examined in Figure 2 are used again. The main oscillator has mass $m_0 = 1000$ kg, damping ratio $\zeta_0 = 0.005$ and natural frequency $\omega_0/(2\pi) = 1$ Hz. The harmonic response $y_0(\omega)$ of the main oscillator without TMD's is characterized by a sharp peak at the resonance frequency, $\omega = \omega_0$. This well-known property is shown in Figure 3, where the amplitude $|y_0(\omega)|$ is normalized by its maximum value $y_{0,\max}$ and plotted as a dash-dot curve.

The harmonic response $y(\omega)$ of the main oscillator with the multiple TMD's is affected by the impedance $Z(\omega)$, which was plotted in Figure 2. For ω near the main oscillator resonance frequency, $\omega_0/(2\pi) = 1$ Hz, the imaginary part of $Z(\omega)$ is relatively small, and the real part is approximately 21 kg/s. According to equation (9), this impedance is equivalent to an increase in the main oscillator damping ratio of about 0.015. Thus, the harmonic response $y(\omega)$ should resemble that of an oscillator with a damping ratio of about 0.02, which would have a resonance peak that is one-fourth of the resonance peak of $y_0(\omega)$. This is shown in Figure 3, where the amplitude $|y(\omega)|$, calculated using the exact expressions (equations (2) and (9)), is normalized by $y_{0,\max}$ and plotted by a solid curve.

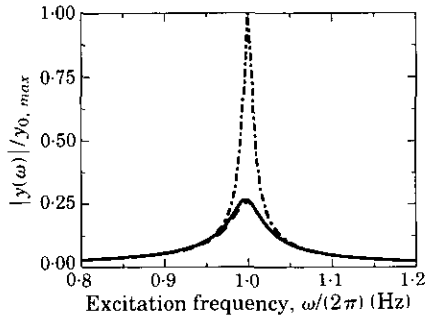


Figure 3. The harmonic response of the main oscillator: ---, without TMD's; —, with TMD's, exact; ···, with TMD's, simple approximation.

The results in Figure 3 show that multiple TMD's with distributed natural frequencies have a damping effect which can be used for passive vibration control. The next step is to determine an optimal design which maximizes the vibration reduction effect. To facilitate this design process, a simple approximation for the harmonic response is needed. Herein, the impedance approximation in equation (7) is used; the expression for the main oscillator harmonic response reduces to

$$y(\omega) \approx \left[m_0(\omega_0^2 - \omega^2 - 2i\omega\omega_0\zeta_0) - \frac{\pi}{2}i\omega^2 p(\omega) \right]^{-1}. \quad (11)$$

To show the accuracy of this simple result, the harmonic response for the system examined in Figure 3 is re-calculated using equation (11). The result, shown in Figure 3 by a dotted curve, nearly coincides with the exact curve, which was determined using the exact, complicated impedance expression in equation (2).

4. OPTIMAL DESIGN OF MULTIPLE TMD'S

To mathematically define an optimal design problem, a performance function and a constraint are necessary. The performance function of interest is the mean-square response of the main oscillator, σ_y^2 , which is given by equation (10). The constraint is the total mass m of the TMD's, which can be computed in terms of the mass density $p(s)$:

$$m = \int_0^\infty [p(\omega)/\omega] d\omega. \quad (12)$$

To make the problem analytically tractable, the approximation in equation (11) is used for the harmonic response. Furthermore, it is assumed that the forcing function can be approximated by white noise, i.e., the input power spectral density is a constant, $G(\omega) = G_0$. This assumption is reasonable for broadband input with smoothly varying power spectral densities [18].

The optimal design of the multiple TMD's is specified by the mass density $p(s)$ which minimizes the integral in equation (10) and satisfies the integral constraint in equation (12). The optimal solution for $p(s)$, which is denoted $p^*(s)$, is derived using calculus of variations. Details of the derivation are given in Appendix A; the results are:

$$p^*(\omega) = (4m_0/\sqrt{3\pi})[\gamma_1 + (2^{4/3} - 1)\sqrt{\gamma_1^2 - (\omega/\omega_0 - 1)^2} - \sqrt{3}\zeta_0], \quad (13)$$

for $\omega_0(1 - \gamma_1) \leq \omega \leq \omega_0(1 + \gamma_1)$, where

$$\gamma_1 = (\pi m/4m_0\zeta_0) \{ \sqrt{1 + (1 + (2^{4/3} - 1)\pi/4)(\pi m/2\sqrt{3}m_0\zeta_0^2) - 1} \}^{-1}. \quad (14)$$

These analytical expressions are simplified by evaluating the numerical constants to three significant digits:

$$p^*(\omega) = 0.735m_0[\gamma_1 + 1.52\sqrt{\gamma_1^2 - (\omega/\omega_0 - 1)^2}] - 1.27m_0\zeta_0, \quad (15)$$

$$\gamma_1 = (0.785m/m_0\zeta_0)(\sqrt{1 + (2m/m_0\zeta_0^2)} - 1)^{-1}. \quad (16)$$

The derivation of $p^*(\omega)$ is independent of the damping ratios of the TMD's, $\zeta(\omega)$. However, there are two conditions on the damping. These conditions, specified by equation (6) and the discussion before equation (7), must be satisfied to use the asymptotic approximations in the preceding section. Using a maximum relative error of $e_m = 10\%$, it was found by numerical comparisons that the conditions are satisfied when the damping ratios of the TMD's lie within the bounds

$$0.77\beta < \zeta(\omega) < \gamma_1. \quad (17)$$

Equations (15)–(17) constitute the general solution to the optimal design problem.

To gain physical insight into the results, the case of lightly damped main oscillators, where $m/m_0 \gg \zeta_0^2$, is examined in detail. This case is not restrictive since a main oscillator with high damping will not need passive control. It can be seen that the second term in the radical in equation (16) becomes dominant, and γ_1 is approximately

$$\gamma_1 \approx 0.556\sqrt{m/m_0}. \quad (18)$$

By substituting this result into equation (15), the following simplified optimal mass density function is obtained:

$$p^*(\omega) = \sqrt{m_0 m} [0.409 + 0.621\sqrt{1 - (3.23m_0/m)(\omega/\omega_0 - 1)^2}], \quad (19)$$

for $\omega_0(1 - 0.556\sqrt{m/m_0}) \leq \omega \leq \omega_0(1 + 0.556\sqrt{m/m_0})$. In Figure 4 it is shown that $p^*(\omega)$ is a half ellipse, with major axis points at $(\omega_0[1 \pm 0.556\sqrt{m/m_0}], 0.409\sqrt{mm_0})$ and one minor axis point at $(\omega_0, 1.03\sqrt{mm_0})$.

The natural frequencies, ω , of the optimally designed TMD's are distributed within $\pm 0.556\omega_0\sqrt{m/m_0}$ of the main oscillator natural frequency, ω_0 . This indicates that multiple TMD's will be less effective if all of their natural frequencies are exactly tuned to ω_0 . A conclusion is that a single TMD will be less effective than the optimally designed multiple TMD's with the same total mass. The mass density of the multiple TMD's is highest when $\omega = \omega_0$ and decreases as the difference between ω and ω_0 increases. The spread of the TMD natural frequencies and the amplitude of the mass density $p^*(\omega)$ are

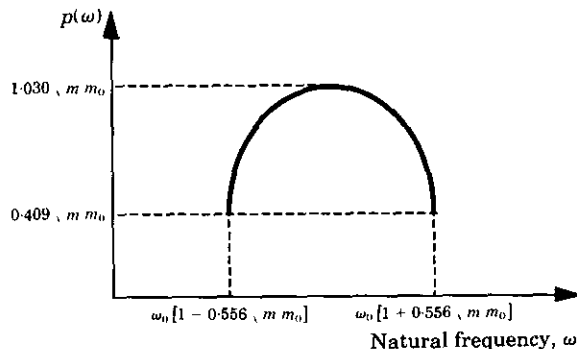


Figure 4. The optimal solution for the mass density function, $p(\omega)$.

both proportional to the square root of the total mass, $\sqrt{m}$. The total area under the curve in Figure 4 is equal to the total mass, m .

To illustrate the results of this section, a main oscillator with properties $m_0 = 1000$ kg, $\zeta_0 = 0.005$ and $\omega_0/(2\pi) = 1$ Hz is considered again. A set of 20 TMD's is used, where each TMD has mass $m_j = 0.5$ kg. This represents a total mass of $m = 10$ kg, which is 1% of the mass of the main oscillator. The optimal mass density is obtained by substituting the parameter values, m_0 , m and ω_0 , into equation (19):

$$p^*(\omega) = [40.9 + 62.1\sqrt{1 - 323(\omega/2\pi - 1)^2}] \text{ kg}, \quad (20)$$

for $0.944 \text{ Hz} \leq \omega/(2\pi) \leq 1.056 \text{ Hz}$. By the definition in equation (3), the mass density is the ratio of the TMD masses over the frequency spacing. Since the masses are already fixed, equation (20) is used to determine the frequency spacing and the natural frequencies of the TMD's. The lowest natural frequency has already been given as $\omega_1/2\pi = 0.944$ Hz. The remaining natural frequencies are specified by $\omega_{j+1} = \omega_j[1 + \beta_j] = \omega_j[1 + m_j/p^*(\omega_j)]$, where $m_j = 0.5$ kg and $p^*(\omega_j)$ is given by equation (20). The results are listed in Table 1. The bounds for the damping ratio in equations (17) and (18) are $0.00939 < \zeta < 0.056$; the value $\zeta_j = 0.05$ is chosen for the example.

Although the mean-square response to wide-band input is of interest, it is informative to examine the harmonic response, $y(\omega)$. Using the parameters of the 20 TMD's given in Table 1 and the exact expressions in equations (4) and (9), the harmonic response is plotted in Figure 5. As in Figure 3, the response is normalized by the resonant response of the oscillator without TMD's, $y_{0,max}$. For comparison, the harmonic response for the main oscillator with a single TMD is also computed, normalized and plotted in Figure 5. The optimal parameters for the single TMD, which were analytically determined by Warburton [4], are used herein. The results show that the harmonic response for the single TMD has two prominent peaks, while that for the multiple TMD's is flatter.

TABLE 1
Properties of multiple TMD's

j	$\omega_j/(2\pi)$ (Hz)	p_j (kg)	β_j
1	0.94436	40.90	0.01223
2	0.95590	78.77	0.00634
3	0.96197	86.23	0.00580
4	0.96755	91.34	0.00547
5	0.97284	95.10	0.00526
6	0.97796	97.92	0.00511
7	0.98295	100.01	0.00450
8	0.98787	101.51	0.00493
9	0.99273	102.47	0.00488
10	0.99758	102.94	0.00486
11	1.00242	102.94	0.00486
12	1.00729	102.46	0.00488
13	1.01221	101.49	0.00493
14	1.01719	99.96	0.00500
15	1.02228	97.80	0.00511
16	1.02751	94.88	0.00527
17	1.03292	90.96	0.00550
18	1.03860	85.63	0.00584
19	1.04467	77.93	0.00642
20	1.05564	40.90	

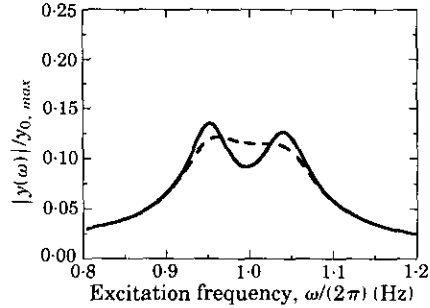


Figure 5. The harmonic response of the main oscillator with optimally designed TMD's: —, single TMD; ---, 20 TMD's.

The root-mean-square response, σ_y , is computed using equation (10) and normalized by the root-mean-square response σ_0 of the oscillator without TMD's. The results are as follows: $\sigma_y/\sigma_0 = 0.408$ for the 20 TMD's and 0.417 for the single TMD. Thus, the example study shows that a main oscillator with optimally designed multiple TMD's has a smoother harmonic response and a slightly lower root-mean-square response than a main oscillator with an optimally designed single TMD.

It is useful to investigate mistuning, where the main oscillator natural frequency differs from the frequency value used in the design of the TMD's. This problem, which is a test of robustness, may arise from modelling or other errors. To evaluate the effectiveness of mistuned TMD's, consider the optimally designed TMD's used in Figure 5, which are designed with the assumption that the main oscillator natural frequency is $\omega_0/(2\pi) = 1$ Hz. The performance of these TMD's is determined for mistuned frequencies, $\omega_0/(2\pi) \neq 1$ Hz.

The normalized harmonic responses for the mistuned case of $\omega_0/(2\pi) = 1.05$ Hz are shown in Figure 6. The harmonic response for the single TMD design has a large peak near 1.05 Hz; in comparison, the response for the multiple TMD design has a slightly lower peak.

The normalized root-mean-square responses for a range of mistuned values for the main oscillator natural frequency are shown in Figure 7. The curves show that, for this problem, multiple TMD's are more effective than a single TMD and are less sensitive to variations of ω_0 . Multiple TMD's are more robust because the natural frequencies of the TMD's are distributed over the range of frequencies shown in Table 1.

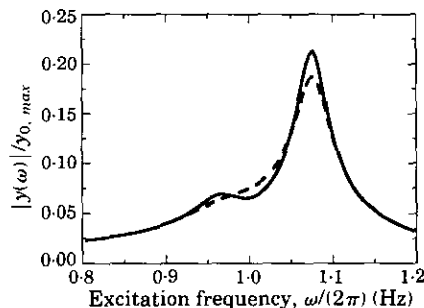


Figure 6. The harmonic response of the main oscillator with mistuned TMD's ($\omega_0 = 1.05$): —, single TMD; ---, 20 TMD's.

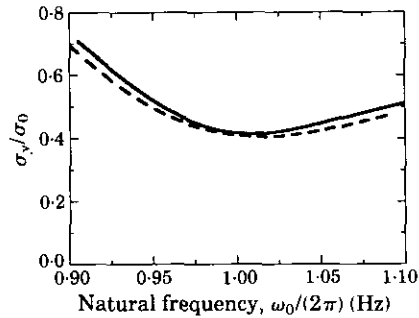


Figure 7. The root-mean-square response of the main oscillator with mistuned natural frequency: —, single TMD; ---, 20 TMD's.

5. SUMMARY AND CONCLUSIONS

This paper has introduced an analysis and design method for multiple TMD's. Several results are presented:

(1) The impedance of multiple oscillators asymptotically approaches an integral form as the natural frequencies become closely spaced. The integral is accurate for moderate numbers of oscillators. It simplifies further when the oscillators are lightly damped.

(2) Multiple TMD's can be optimally designed for wide-band input using the calculus of variations. The simple form for the impedance approximation allows for a closed form expression for the design parameters. The optimal result shows that the natural frequencies of the TMD's should be distributed over a frequency range centered at the main oscillator natural frequency. The width of this frequency range is proportional to the square root of the total mass of the TMD's.

(3) Optimally designed multiple TMD's are more effective and more robust than an optimally designed single TMD of equal total mass. In particular, the main oscillator with multiple TMD's has a flatter harmonic response and a lower root-mean-square response than the oscillator with a single TMD.

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APPENDIX A: DETAILS IN SOLVING THE OPTIMAL DESIGN PROBLEM

Calculus of variations is used to derive the optimal function for $p(\omega)$ which minimizes the integral in equation (10) and satisfies the integral constraint in equation (12). The basic procedure [16] is as follows. The Hamiltonian $H(p, \omega)$ is determined, and the Euler–Lagrange equation, $\partial H/\partial p = 0$ is formulated. Then, the Euler–Lagrange and constraint equations are solved simultaneously to obtain the optimal function for $p(\omega)$.

The Hamiltonian is

$$H(p, \omega) = G_0 |y(\omega)|^2 + \lambda p(\omega)/\omega \\ = (G_0/m_0^2) \{(\omega^2 - \omega_0^2)^2 + 4[\omega\omega_0\zeta_0 + (\pi/4)\omega^2 p(\omega)]^2\}^{-1} + \lambda p(\omega)/\omega, \quad (\text{A1})$$

where λ is the Lagrange multiplier. This function is simplified by introducing three non-dimensional parameters and one assumption. The parameters are

$$\gamma = (\omega - \omega_0)/\omega_0, \quad \theta = 2\zeta_0 + \pi p(\omega)/2m_0, \quad \lambda' = 2m_0^2\omega_0^3\lambda/\pi G_0. \quad (\text{A2})$$

The assumption is that the natural frequencies of the multiple TMD's will be near that of the main oscillator, i.e., $\omega \approx \omega_0$ or $\gamma \ll 1$. This assumption is reasonable since TMD's become ineffective when their natural frequencies become far from that of the main oscillator. By substituting equation (A2) into equation (A1) and retaining lowest order terms in γ , the following simplified form for the Hamiltonian is obtained:

$$H(\theta, \gamma) \approx (G_0/m_0^2\omega_0^4) [1 + \lambda'(\theta - 2\zeta_0)(4\gamma^2 + \theta^2)]/(4\gamma^2 + \theta^2). \quad (\text{A3})$$

The Euler–Lagrange equation for the optimization problem is $\partial H/\partial \theta = 0$, which, for the Hamiltonian in equation (A3), becomes

$$\lambda'(4\gamma^2 + \theta^2)^2 - 2\theta = 0, \quad (\text{A4})$$

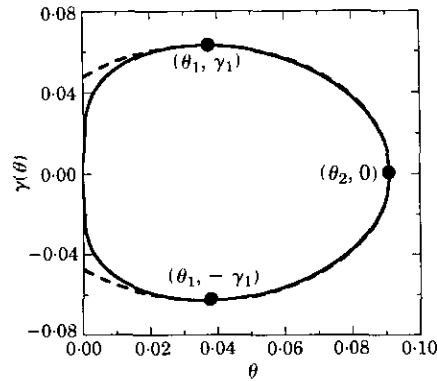


Figure A1. A graph of γ vs. θ : ---, exact (equation (A5)); —, elliptical approximation (equation (A7)).

An explicit analytical solution can be obtained for γ in terms of θ and λ' :

$$\gamma = \pm 2\sqrt{\sqrt{2\theta/\lambda'} - \theta^2}. \quad (\text{A5})$$

Since γ must be real-valued, the positive root is taken for the inner radical.

To obtain an explicit expression for θ in terms of γ and λ' , it is helpful to examine the relationship between θ and γ . The first step is to determine the co-ordinate pairs (θ_1, γ_1) and (θ_2, γ_2) which correspond to the maximum values of γ and θ , respectively. The first pair of co-ordinates is determined by differentiating equation (A5) with respect to θ and setting the result to 0. The second co-ordinate pair is the non-trivial solution to $\gamma = 0$. The results are

$$(\theta_1, \gamma_1) = (1/2 \sqrt[3]{\lambda'}, \sqrt{3/\sqrt[3]{\lambda'}}), \quad (\theta_2, \gamma_2) = (\sqrt[3]{2/\lambda'}, 0). \quad (\text{A6})$$

A typical graph of γ versus θ is shown in Figure A1 (dashed line) for $\lambda' = 2618$. The graph is symmetric with respect to the horizontal axis. The graph also shows that for $|\gamma| < \gamma_1$, there are two values of θ , and for $|\gamma| > \gamma_1$, there are no values for θ . A physical interpretation is gained by examining the definitions for γ and θ in equation (A2). The frequency parameter, γ , measures the separation between the natural frequency of a TMD and that of the main oscillator. The design parameter, θ , is linearly related to the mass density of the multiple TMD's. The fact that there are no values for θ for $|\gamma| > \gamma_1$ implies that there should be no TMD's with natural frequencies sufficiently far from that of the main oscillator. Of the two values for θ which appear for $|\gamma| < \gamma_1$, the smaller value can be excluded. This is because it represents a non-optimal design solution in which the multiple TMD's approach zero mass density as the corresponding natural frequencies approach ω_0 .

The preceding discussion shows that the relationship between γ and θ is given by the portion of the curve through the points (θ_1, γ_1) , $(\theta_2, 0)$ and $(\theta_1, -\gamma_1)$. This curve can be approximated by the right half of an ellipse with one major axis point at $(\theta_2, 0)$ and minor axis points at $(\theta_1, \pm \gamma_1)$. A comparison between the exact curve (equation (A5)) and the ellipse is shown in Figure A1. Numerical tests show that the ellipse approximation is accurate for all practical values of λ' . The equation for the ellipse gives the explicit expression for θ in terms of γ and λ' :

$$\theta = \theta_1 + (\theta_2 - \theta_1)\sqrt{1 - (\gamma/\gamma_1)^2}, \quad (\text{A7})$$

for $|\gamma| < \gamma_1$.

The constraint equation can be solved analytically using the preceding results. First, equations (A2) and (A7) are substituted into equation (12) to yield

$$m \approx (2m_0/\pi) \left(\int_{-\gamma_1}^{\gamma_1} \theta(\gamma) d\gamma - 4\zeta_0\gamma_1 \right). \quad (\text{A8})$$

The integral in equation (A8) is equal to the area between the half-ellipse and the vertical axis in Figure A1. From simple geometry, it can be shown that this area is equal to $2\gamma_1\theta_1 + \pi\gamma_1(\theta_2 - \theta_1)/2$. By combining these results with equation (A6), a solution for the normalized Lagrange multiplier is obtained:

$$\lambda' = \frac{1}{64} \{ (m_0\zeta_0/\pi m) [(48 + 2\sqrt{3}[4 + (2^{4/3} - 1)\pi](\pi m/m_0\zeta_0^2))^{1/2} - 4\sqrt{3}] \}^3. \quad (\text{A9})$$

The final expression for the optimal design of the multiple TMD's is obtained by reformulating equations (A6) and (A7) in terms of the original physical parameters of the problem. The results are given in equations (13) and (14) in the main text.